

Reckitt Sales Analysis

Robert Science Data Analytics Consulting

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Introducción

En este proyecto trabajé directamente con datos reales de ventas, enfocándome en construir una base de datos desde cero, limpiarla y dejarla lista para análisis. La idea no fue solo cargar información, sino asegurar que los datos fueran consistentes, confiables y útiles para la toma de decisiones.

Durante el desarrollo, me enfoqué en estructurar bien el flujo de trabajo: desde la creación de tablas, la carga de datos, hasta la validación y análisis. Todo el proceso lo trabajé utilizando SQL Server, apoyándome en Visual Studio Code para la gestión del código y herramientas como pandas para validaciones externas cuando fue necesario.

Capturas del proceso

entregable_4.sql

```
sql> entregable_4.sql
localhost\SQLEXPRESS | Reckitt_Sales_Analysis
USE master;
GO

IF DB_ID('Reckitt_Sales_Analysis') IS NOT NULL
BEGIN
    ALTER DATABASE Reckitt_Sales_Analysis
    SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE Reckitt_Sales_Analysis;
END
GO

CREATE DATABASE Reckitt_Sales_Analysis;
GO

USE Reckitt_Sales_Analysis;
GO

-- =====
-- 1. DIMENSIONES
-- =====

CREATE TABLE DIM_CATEGORY (
    ID_CATEGORY NVARCHAR(50),
    CATEGORY NVARCHAR(200)
);

CREATE TABLE DIM_SEGMENT (
    ID_SEGMENT NVARCHAR(50),
    SEGMENT NVARCHAR(200)
);
```

Results (14) Messages

TYPE	ROWS
RAW	122002
CLEAN	122002

TOTAL_ROWS	NULL_VALUES
122002	0

MATCHED_ROWS
122002

UNMATCHED_ROWS
0

REGION	TOTAL_SALES
TOTAL AUTOS SCANNING MEXICO	5521429.320000029
TOTAL AUTOS AREA 2	1188796.1500000011
TOTAL AUTOS AREA 5	1153335.538000006
TOTAL AUTOS AREA 6	983957.5709999978
TOTAL AUTOS AREA 3	803655.3370000044

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 13, Col 3 Spaces: 4 UTF-8 CRLF () SQL Finish Setup Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost\SQLEXPRESS Reckitt_Sales_Analysis

entregable_4.sql

```
sql> entregable_4.sql
localhost\SQLEXPRESS | Reckitt_Sales_Analysis
USE master;
GO

IF DB_ID('Reckitt_Sales_Analysis') IS NOT NULL
BEGIN
    ALTER DATABASE Reckitt_Sales_Analysis
    SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE Reckitt_Sales_Analysis;
END
GO

CREATE DATABASE Reckitt_Sales_Analysis;
GO

USE Reckitt_Sales_Analysis;
GO

-- =====
-- 1. DIMENSIONES
-- =====

CREATE TABLE DIM_CATEGORY (
    ID_CATEGORY NVARCHAR(50),
    CATEGORY NVARCHAR(200)
);

CREATE TABLE DIM_SEGMENT (
    ID_SEGMENT NVARCHAR(50),
    SEGMENT NVARCHAR(200)
);

CREATE TABLE DIM_PRODUCT (
    MANUFACTURER NVARCHAR(100),
    BRAND NVARCHAR(100),
    ITEM NVARCHAR(50),
    ITEM_DESCRIPTION NVARCHAR(300),
    CATEGORY NVARCHAR(100),
    FORMAT NVARCHAR(50),
    ATTR1 NVARCHAR(100),
    ATTR2 NVARCHAR(100),
    ATTR3 NVARCHAR(100)
);

CREATE TABLE DIM_CALENDAR (
    WEEK NVARCHAR(50),
    YEAR NVARCHAR(50),
    MONTH NVARCHAR(50),
    WEEK_NUMBER NVARCHAR(50),
    DATE_VALUE NVARCHAR(50)
);
```

Results (14) Messages

ITEM_CODE	TOTAL_SALES
0000075000639	1146849.6320000002
0000075000622	1139233.787
0000075000615	658584.5679999996
7501025402051	491075.23199999996
7501071900143	399994.77899999986

CATEGORY_NAME	TOTAL_SALES
"FABRIC TREATMENT and SANIT"	11042859.892999997

WEEK	TOTAL_SALES
"01-22"	151538.92200000008
"01-23"	166309.04399999999
"02-22"	146996.99099999978
"02-23"	155655.88199999998

PS D:\Documentos\Ebac\Finales\entregable_4>
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Ln 13, Col 3 Spaces: 4 UTF-8 CRLF () SQL Finish Setup Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost\SQLEXPRESS Reckitt_Sales_Analysis

entregable_4

entregable_4.sql

```
sql> CREATE TABLE DIM_CALENDAR (
51     MONTH NVARCHAR(50),
52     WEEK_NUMBER NVARCHAR(50),
53     DATE_VALUE NVARCHAR(50)
54 );
55 GO
56
57 PRINT 'DIMENSIONS CREATED';
58 GO
59
60 -- =====
61 -- 2. FACT RAW
62 -- =====
63
64 CREATE TABLE FACT_SALES_RAW (
65     WEEK NVARCHAR(50),
66     ITEM_CODE NVARCHAR(50),
67     TOTAL_UNIT_SALES NVARCHAR(50),
68     TOTAL_VALUE_SALES NVARCHAR(50),
69     TOTAL_UNIT_AVG_WEEKLY_SALES NVARCHAR(50),
70     REGION NVARCHAR(100)
71 );
72 GO
73
74 PRINT 'RAW TABLE CREATED';
75 GO
76
77 -- =====
```

Results (14) Messages

WEEK	TOTAL_SALES
"02-22"	146996.99099999978
"02-23"	155655.88199999998
"03-22"	125351.33600000045
"03-23"	135774.212
"04-22"	123945.61699999975
"04-23"	126281.49099999995
"05-22"	129841.76700000023
"22-23"	199783.53399999993

AVG_PRICE
28.18780239478688

ITEM_CODE	PRICE_PER_UNIT
7501080900585	262.1420108709261
7501058795908	251.78227408324992
7501058715883	238.4808115668738
7501058715890	233.90513844307
7501080952997	228.8389308774216

Build with Agent

AI responses may be inaccurate

Generate Agent

Instructions to onboard AI onto your codebase.

Tip: Try the Plan agent to research and plan before implementing changes.

+ entregable_4.sql

Describe what to build

+ ...

+ ...

+ ...

PS D:\Documentos\Ebac\Finales\entregable_4>

PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 13, Col 3 Spaces: 4 UTF-8 CRLF () SQL Finish Setup (Go) Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost/SQLEXPRESS Reckitt_Sales_Analysis

entregable_4

entregable_4.sql

```
sql> PRINT 'RAW TABLE CREATED';
75 GO
76
77 -- =====
78 -- 3. LOAD DATA
79 -- =====
80
81 BULK INSERT DIM_CATEGORY
82 FROM 'D:\Documentos\Ebac\Finales\entregable_4\data
83 WITH (FIRSTROW=2, FIELDTERMINATOR=',', ROWTERMINAT
84
85 BULK INSERT DIM_SEGMENT
86 FROM 'D:\Documentos\Ebac\Finales\entregable_4\data
87 WITH (FIRSTROW=2, FIELDTERMINATOR=',', ROWTERMINAT
88
89 BULK INSERT DIM_PRODUCT
90 FROM 'D:\Documentos\Ebac\Finales\entregable_4\data
91 WITH (FIRSTROW=2, FIELDTERMINATOR=',', ROWTERMINAT
92
93 BULK INSERT DIM_CALENDAR
94 FROM 'D:\Documentos\Ebac\Finales\entregable_4\data
95 WITH (FIRSTROW=2, FIELDTERMINATOR=',', ROWTERMINAT
96
97 BULK INSERT FACT_SALES_RAW
98 FROM 'D:\Documentos\Ebac\Finales\entregable_4\data
99 WITH (FIRSTROW=2, FIELDTERMINATOR=',', ROWTERMINAT
100 GO
101
```

Results (14) Messages

CATEGORY	TOTAL_SALES
"FABRIC TREATMENT and SANIT"	11042859.892999999

CATEGORY_NAME	TOTAL_SALES
"FABRIC TREATMENT and SANIT"	11042859.892999999

CATEGORY_NAME	REGION
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS SCANNI
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA
"FABRIC TREATMENT and SANIT"	TOTAL AUTOS AREA

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File Edit Selection View Go Run ...

entregable_4

Update

entregable_4.sql 4 X clean_all_csv.py clean_fact_sales.py load_to_sql.py 2 convert_all.py

EXPLORER

ENTRE...
data
DIM_CALENDAR.csv
DIM_CALENDAR.xlsx
DIM_CATEGORY.csv
DIM_PRODUCT.csv
DIM_PRODUCT.xlsx
DIM_SEGMENT.csv
DIM_SEGMENT.xlsx
FACT_SALES.csv
docs
capturas
000.png
001.png
002.png
003.png
README.md
sql
clean_fact_sales.py
entregable_4.sql 4
clean_all_csv.py
convert_all.py
convert_excel.py
load_to_sql.py 2
reporte_entregable4.h...

OUTLINE
TIMELINE

sql > entregable_4.sql

100 GO
101
102 PRINT 'DATA LOADED';
103 GO
104
105 -- =====
106 -- 4. CLEAN LAYER
107 -- =====
108
109 DROP TABLE IF EXISTS FACT_SALES;
110 GO
111
112 SELECT
113 LTRIM(RTRIM(WEEK)) AS WEEK,
114
115 REPLACE(REPLACE(LTRIM(RTRIM(ITEM_CODE)),' ',''),',','') AS ITEM_CODE,
116
117 TRY_CONVERT(FLOAT, REPLACE(REPLACE(TOTAL_UNIT_SALES,' ',''),',','')) AS TOTAL_UNIT_SALES,
118 TRY_CONVERT(FLOAT, REPLACE(REPLACE(TOTAL_VALUE_SALES,' ',''),',','')) AS TOTAL_VALUE_SALES,
119 TRY_CONVERT(FLOAT, REPLACE(REPLACE(TOTAL_UNIT_AVG_WEEKLY_SALES,' ',''),',','')) AS AVG_WEEKLY_SALE
120
121 LTRIM(RTRIM(REGION)) AS REGION
122 INTO FACT_SALES
123 FROM FACT_SALES_RAW;
124 GO
125
126 PRINT 'CLEAN FACT CREATED';
127 GO
128

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4> |
Ln 13, Col 3 Spaces: 4 UTF-8 CRLF () SQL Finish Setup Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost\SQLEXPRESS Reckitt_Sales_Analysis

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+ entregable_4.sql
Describe what to build
+ ...

File Edit Selection View Go Run ...

entregable_4

Update

entregable_4.sql 4 X clean_all_csv.py clean_fact_sales.py load_to_sql.py 2 convert_all.py

EXPLORER

ENTRE...
data
DIM_CALENDAR.csv
DIM_CALENDAR.xlsx
DIM_CATEGORY.csv
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DIM_SEGMENT.xlsx
FACT_SALES.csv
docs
capturas
000.png
001.png
002.png
003.png
004.png
README.md
sql
clean_fact_sales.py
entregable_4.sql 4
clean_all_csv.py
convert_all.py
convert_excel.py
load_to_sql.py 2
reporte_entregable4.h...

OUTLINE
TIMELINE

sql > entregable_4.sql

129 -- =====
130 -- 5. VALIDACIÓN
131 -- =====
132
133 SELECT 'RAW' AS TYPE, COUNT(*) AS ROWS FROM FACT_SALES_RAW
134 UNION ALL
135 SELECT 'CLEAN', COUNT(*) FROM FACT_SALES;
136 GO
137
138 -- =====
139 -- 6. DATA QUALITY CHECK
140 -- =====
141
142 -- NULOS
143 SELECT
144 COUNT(*) AS TOTAL_ROWS,
145 SUM(CASE WHEN TOTAL_VALUE_SALES IS NULL THEN 1 ELSE 0 END) AS NULL_VALUES
146 FROM FACT_SALES;
147 GO
148
149 -- MATCH DE JOIN
150 SELECT
151 COUNT(*) AS MATCHED_ROWS
152 FROM FACT_SALES f
153 JOIN DIM_PRODUCT p
154 ON REPLACE(f.ITEM_CODE,' ','') = REPLACE(p.ITEM,' ','');
155 GO
156

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
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+ entregable_4.sql
Describe what to build
+ ...

entregable_4

Update

EXPLORER

- ENTREGABLE_4
 - data
 - DIM_CALENDAR.csv
 - DIM_CALENDAR.xlsx
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 - DIM_PRODUCT.csv
 - DIM_PRODUCT.xlsx
 - DIM_SEGMENT.csv
 - DIM_SEGMENT.xlsx
 - FACT_SALES.csv
 - docs
 - capturas
 - 000.png
 - 001.png
 - 002.png
 - 003.png
 - 004.png
 - 005.png
 - README.md
 - sql
 - clean_fact_sales.py
 - entregable_4.sql 4
 - clean_all_csv.py
 - convert_all.py
 - convert_excel.py
 - load_to_sql.py 2
 - reporte_entregable4.h...
 - OUTLINE
 - TIMELINE

```
sql > entregable_4.sql
156
157 -- NO MATCH
158 SELECT COUNT(*) AS UNMATCHED_ROWS
159 FROM FACT_SALES f
160 LEFT JOIN DIM_PRODUCT p
161     ON REPLACE(f.ITEM_CODE, '', '') = REPLACE(p.ITEM, '', '')
162 WHERE p.ITEM IS NULL;
163 GO
164
165 -- =====
166 -- 7. ANALYTICS
167 -- =====
168
169 -- VENTAS POR REGIÓN
170 SELECT
171     REGION,
172     SUM(TOTAL_VALUE_SALES) AS TOTAL_SALES
173 FROM FACT_SALES
174 GROUP BY REGION
175 ORDER BY TOTAL_SALES DESC;
176 GO
177
178 -- TOP PRODUCTOS
179 SELECT TOP 5
180     ITEM_CODE,
181     SUM(TOTAL_VALUE_SALES) AS TOTAL_SALES
182 FROM FACT_SALES
183 GROUP BY ITEM_CODE
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 13, Col 3 Spaces: 4 UTF-8 CRLF () SQL Finish Setup Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost/SQLEXPRESS Reckitt_Sales_Analysis

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+ entregable_4.sql

Describe what to build

+ ...

entregable_4

Update

EXPLORER

- ENTREGABLE_4
 - data
 - DIM_CALENDAR.csv
 - DIM_CALENDAR.xlsx
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 - DIM_SEGMENT.csv
 - DIM_SEGMENT.xlsx
 - FACT_SALES.csv
 - docs
 - capturas
 - 000.png
 - 001.png
 - 002.png
 - 003.png
 - 004.png
 - 005.png
 - 006.png
 - README.md
 - sql
 - clean_fact_sales.py
 - entregable_4.sql 4
 - clean_all_csv.py
 - convert_all.py
 - convert_excel.py
 - load_to_sql.py 2
 - OUTLINE
 - TIMELINE

```
sql > entregable_4.sql
182 FROM FACT_SALES
183 GROUP BY ITEM_CODE
184 ORDER BY TOTAL_SALES DESC;
185 GO
186
187 -- VENTAS POR CATEGORÍA
188 SELECT
189     c.CATEGORY AS CATEGORY_NAME,
190     SUM(f.TOTAL_VALUE_SALES) AS TOTAL_SALES
191 FROM FACT_SALES f
192 JOIN DIM_PRODUCT p
193     ON REPLACE(f.ITEM_CODE, '', '') = REPLACE(p.ITEM, '', '')
194 JOIN DIM_CATEGORY c
195     ON p.CATEGORY = c.ID_CATEGORY
196 GROUP BY c.CATEGORY
197 ORDER BY TOTAL_SALES DESC;
198 GO
199
200 -- =====
201 -- 8. ANÁLISIS TEMPORAL
202 -- =====
203
204 SELECT
205     WEEK,
206     SUM(TOTAL_VALUE_SALES) AS TOTAL_SALES
207 FROM FACT_SALES
208 GROUP BY WEEK
209 ORDER BY WEEK;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 204, Col 8 Spaces: 4 UTF-8 CRLF () SQL Finish Setup Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost/SQLEXPRESS Reckitt_Sales_Analysis

Build with Agent

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+ entregable_4.sql

Describe what to build

+ ...

entregable_4

File Edit Selection View Go Run ...

entregable_4.sql 4 • clean_all_csv.py • clean_fact_sales.py • load_to_sql.py 2 • convert_all.

ENTREGABLE_4

- data
 - DIM_CALENDAR.csv
 - DIM_CALENDAR.xlsx
 - DIM_CATEGORY.csv
 - DIM_PRODUCT.csv
 - DIM_PRODUCT.xlsx
 - DIM_SEGMENT.csv
 - DIM_SEGMENT.xlsx
 - FACT_SALES.csv
- docs
 - capturas
 - 000.png
 - 001.png
 - 002.png
 - 003.png
 - 004.png
 - 005.png
 - 006.png
 - 007.png
 - README.md
 - sql
 - clean_fact_sales.py
 - entregable_4.sql 4
 - clean_all_csv.py
 - convert_all.py
 - convert_excel.py
- OUTLINE
- TIMELINE

```
sql > entregable_4.sql
208 GROUP BY WEEK
209 ORDER BY WEEK;
210 GO
211
212 SELECT TOP 1
213     WEEK,
214     SUM(TOTAL_VALUE_SALES) AS TOTAL_SALES
215 FROM FACT_SALES
216 GROUP BY WEEK
217 ORDER BY TOTAL_SALES DESC;
218 GO
219
220 -- =====
221 -- 9. KPI
222 -- =====
223
224 -- PRECIO PROMEDIO
225 SELECT
226     SUM(TOTAL_VALUE_SALES) / NULLIF(SUM(TOTAL_UNIT_SALES),0) AS AVG_PRICE
227 FROM FACT_SALES;
228 GO
229
230 -- PRECIO POR PRODUCTO
231 SELECT TOP 10
232     ITEM_CODE,
233     SUM(TOTAL_VALUE_SALES) / NULLIF(SUM(TOTAL_UNIT_SALES),0) AS PRICE_PER_UNIT
234 FROM FACT_SALES
235 GROUP BY ITEM_CODE
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 204, Col 8 Spaces: 4 UTF-8 CRLF () SQL Finish Setup (Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost\SQLEXPRESS Reckitt_Sales_Analysis

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+ entregable_4.sql

Describe what to build

+ ...

entregable_4

File Edit Selection View Go Run ...

entregable_4.sql 4 • clean_all_csv.py • clean_fact_sales.py • load_to_sql.py 2 • convert_all.

ENTREGABLE_4

- data
 - DIM_CALENDAR.csv
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 - capturas
 - 000.png
 - 001.png
 - 002.png
 - 003.png
 - 004.png
 - 005.png
 - 006.png
 - 007.png
 - 008.png
 - README.md
 - sql
 - clean_fact_sales.py
 - entregable_4.sql 4
 - clean_all_csv.py
 - convert_all.py
- OUTLINE
- TIMELINE

```
sql > entregable_4.sql
234 FROM FACT_SALES
235 GROUP BY ITEM_CODE
236 ORDER BY PRICE_PER_UNIT DESC;
237 GO
238
239 -- =====
240 -- 10. VISTA ANALÍTICA
241 -- =====
242
243 CREATE OR ALTER VIEW VW_SALES_ANALYSIS AS
244 SELECT |
245     f.WEEK,
246     f.REGION,
247     p.BRAND,
248     c.CATEGORY,
249     f.TOTAL_VALUE_SALES,
250     f.TOTAL_UNIT_SALES
251 FROM FACT_SALES f
252 JOIN DIM_PRODUCT p
253     ON REPLACE(f.ITEM_CODE,' ','') = REPLACE(p.ITEM,' ','')
254 JOIN DIM_CATEGORY c
255     ON p.CATEGORY = c.ID_CATEGORY;
256 GO
257
258 -- USO DE LA VISTA
259 SELECT
260     CATEGORY,
261     SUM(TOTAL_VALUE_SALES) AS TOTAL_SALES
262 FROM VW_SALES_ANALYSIS
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Documentos\Ebac\Finales\entregable_4>
PS D:\Documentos\Ebac\Finales\entregable_4>

Ln 244, Col 8 Spaces: 4 UTF-8 CRLF () SQL Finish Setup (Go Live MSSQL SQLCMD: Off 7 rows affected 00:07:46.212 localhost\SQLEXPRESS Reckitt_Sales_Analysis

Build with Agent

AI responses may be inaccurate

Generate Agent

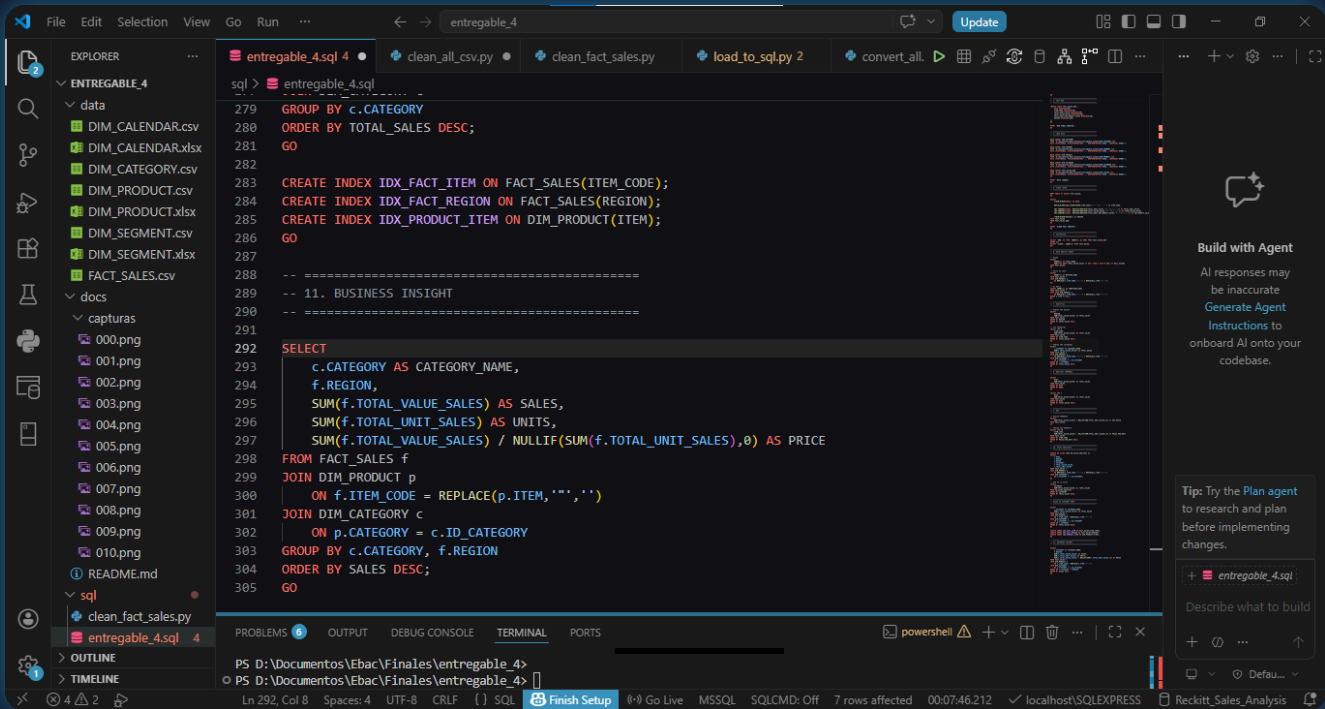
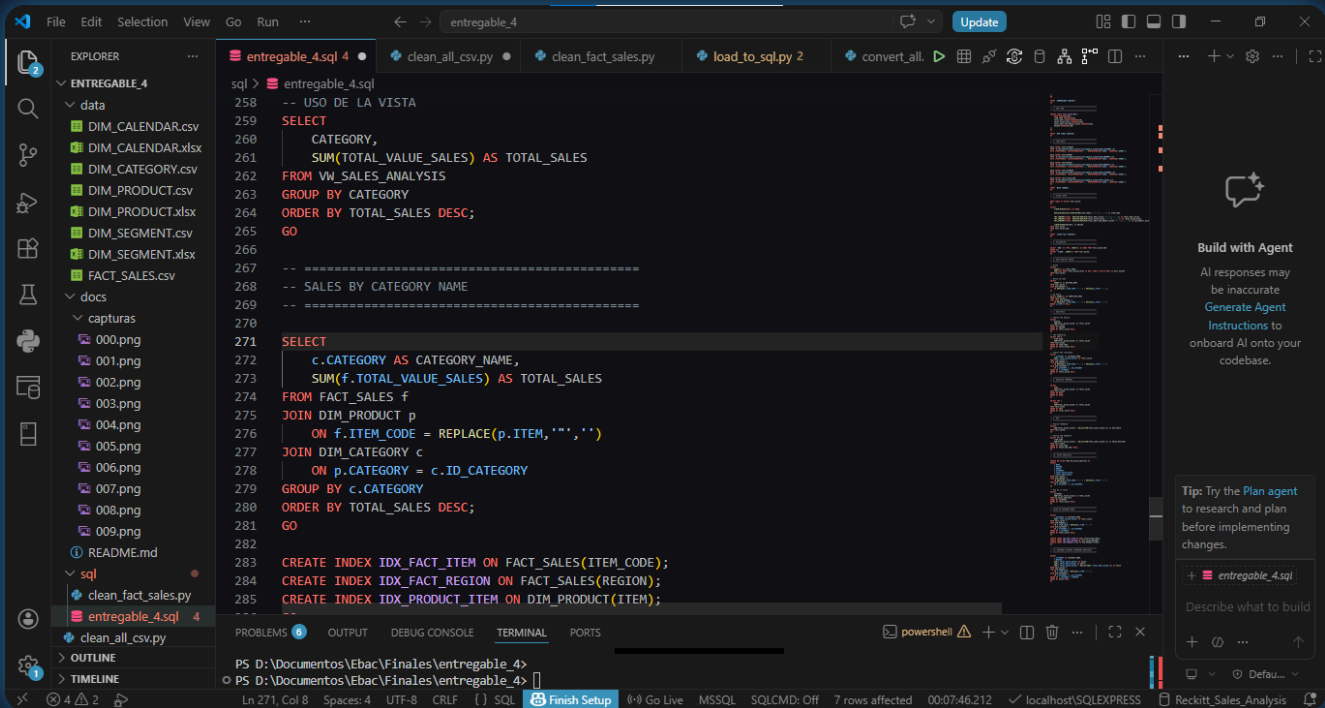
Instructions to onboard AI onto your codebase.

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+ entregable_4.sql

Describe what to build

+ ...



Desarrollo del proyecto

Primero construí la base de datos completa, definiendo tablas de dimensiones y una tabla de hechos en formato raw para poder controlar el proceso de limpieza. Posteriormente cargué los datos utilizando BULK INSERT, asegurando que todos los archivos se integraran correctamente.

Después trabajé en la limpieza de los datos, eliminando caracteres innecesarios, corrigiendo formatos numéricos y asegurando que las llaves coincidieran correctamente entre tablas. Este paso fue clave para evitar errores en los joins.

Una vez limpia la información, validé que no existieran pérdidas de datos ni inconsistencias. Comparé registros entre tablas raw y limpias, revisé valores nulos y confirmé que el 100% de los registros coincidieran en los cruces entre tablas.

Finalmente desarrollé distintos análisis: ventas por región, productos más vendidos, comportamiento por categoría y evolución temporal. También calculé indicadores clave como el precio promedio y el precio por producto, lo que permite entender mejor el valor de cada segmento.

Conclusión

El resultado fue una base de datos completamente funcional, limpia y lista para análisis. Se logró mantener la integridad de los datos en todo momento, asegurando consistencia en los resultados y permitiendo generar insights claros sobre el comportamiento de las ventas.

Este tipo de trabajo demuestra la importancia de un buen proceso de limpieza y validación, ya que cualquier error en esta etapa impacta directamente en los análisis posteriores.

Aprendizajes

Durante este proyecto reforcé la importancia de estructurar bien los datos desde el inicio. Aprendí a trabajar con información en bruto, transformarla y validar cada paso para asegurar calidad.

También entendí mejor cómo conectar distintas tablas correctamente y cómo pequeños detalles, como limpiar strings o formatos, pueden hacer una gran diferencia en los resultados.

Además, este proyecto me permitió trabajar de una forma más cercana a un entorno real, pensando no solo en que el código funcione, sino en que sea claro, ordenado y útil para análisis de negocio.

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